

Pay It Forward

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Let me ask you a question: what experience got you most excited to enter the field you're in now? What got you excited about coding or electronics? For me, I have vivid memories of playing with all kinds of science kits as a young kid: chemistry sets, rock tumblers, and probably six different crystal radios. And it was the electronics that grabbed my attention the most. I remember being in awe of the things you could do with electronics (Blink lights! Make sounds! Build robots!), and all of it following clearly identifiable rules (Ohm's Law, Kirchhoff's Laws, etc). With a little bit of work, I could understand the world around me, I could predict what it would do, nay, I could even control it and make it do what I wanted it to do!

However, not everyone around the world today is as blessed as you or I were and many children who would otherwise love and excel in a tech-related career aren't given that chance, simply for lack of those experiences that you and I had when we were youngsters. But the good news is that you can help change that! You can join in the work of others to share the joy and wonder of coding and electronics with kids around the world. You can be the person who ignites their passion and helps them believe that they are capable of more than they ever dared dream. It can be small, at first; even donating a few dollars a month to a worthy cause can help sustain their important missions. Yours may very well be the spark that starts the fire. Later on, by volunteering with your time, perhaps you could be the person who inspires a youngster much like you may have been encouraged and inspired when you were young.

If you can already think of a cause or organization to support, that's great! Go with that! If not, here is a short list of organizations that I think are pretty cool, several of which I donate to already myself. Which one of them fires you up and makes you want to jump in and help make a difference?

Digital NEST

You can give:



Digital NEST is an organization whose mission is to help (primarily Latin) youth start promising careers in various tech fields. They do this by forming and running learning centers where youth can develop the technical and soft skills needed for their chosen careers; there are currently five locations, all in Northern California. To date, they've helped over 4,000 youth develop the skills to get hired at places such as Superset, Adobe, KSBW-TV, and the Monterey Bay Aquarium.

In addition to donating money, you can support Digital NEST by:

- Hiring a Digital NEST intern
- Becoming a Digital NEST mentor
- Contracting with bizzNEST, Digital NEST's in-house digital media agency, for web development, video production, and graphic design

Elevate Tutoring

You can give:



[Elevate Tutoring](#) is an organization (founded by one of my undergraduate professors!) that helps students of all backgrounds overcome financial and achievement gaps to enroll and succeed in college. They do this through a sort of positive feedback loop, where they provide "training, work experience, career development, and scholarships to low-income, first-generation college students - our Tutor Fellows - who, in turn, provide high-quality STEM tutoring, mentorship, and growth mindset coaching to their younger K-12 peers" ... who then become Tutor Fellows themselves, thus continuing the cycle!

In addition to donating money, you can support Elevate Tutoring by:

- Becoming an Elevate Community Advisor
- Attending an Elevate Tutoring event
- Spreading the word about Elevate Tutoring's mission, particularly to current college students that you think could benefit from participating in their program (see the eligibility [here](#) for additional details).

Pass the Bricks

You can give:



[Pass the Bricks](#) is an organization founded in 2020 by Charlie Jeffers, a (then) high school student from Marin County, CA, whose mission is to distribute LEGO kits to kids who can't afford them. The organization accepts donations of LEGO bricks, creates their own kits out of the parts, and then sends them out to kids in need! Chances are, you've played with LEGOs as a kid and can remember the thrill of being able to create just about anything that came to your imagination. Ralph Hempel, in [episode 466 of the Embedded.fm podcast](#), made the great point that "I think if you sampled ... any engineers that you know, most of them will have had Lego in their hands at some point in their lives. It is not that playing with Lego makes you an engineer, but most engineers ... in my experience have touched Lego."

In addition to donating your own old LEGO kits to Pass the Bricks, you can also volunteer:

- to assist in the process of cleaning and repackaging the kits that are sent out or
- to be an "ambassador" for Pass the Bricks.

Embedded Artistry

You can give:



[Embedded Artistry](#) is a website and contracting business run by Phillip Johnston. His website has an unbelievable amount of information about embedded systems, some of which is available to members only. By [sponsoring a membership for another person](#), you could provide access to this incredible resource to individuals around the world who might not have the financial means to access it otherwise.

Girls Who Code

You can give:



Girls Who Code is an organization whose mission is to close the gender gap in tech. To date, they've taught over 670,000 girls, women, and non-binary individuals, and with tremendous success: Girls Who Code alumni are earning college degrees in computer science and related fields at seven times the national average!

In addition to donating money, you can support Girls Who Code by:

- Starting a GWC club
- Fundraising for GWC
- Participating in a GWC Campaign
- Letting your local school or your friends know about GWC, so they can participate in their awesome programming!

Donor's Choose

You can give:



[Donor's Choose](#) is an organization founded in 2000 by a public school teacher in the Bronx to help fulfill the gap left by government funding. Public school teachers around the US can post a "wish list" of items they would like to have for their classrooms and donors can fund these requests specifically or in general. You can also use their website to search by project or by zip code. I wonder if there's a teacher at a school right around the corner from you that earnestly needs basic school supplies! Or maybe you can help just by letting your local school know that the site exists[®], so that others can help fulfill their needs.

Hack Club

You can give:



[Hack Club](#) is an on- and off-line group of teenage hackers from around the world whose goal is to create a supportive learning community for folks like them. Their members have been responsible for such awesome projects as [Sine Rider](#) (an unbelievably beautiful and fun online game to teach people about graphing functions) and [Some Assembly Required](#). They send out [free electronics kits](#), they'll help you [build your first PCB](#) or [handheld game](#), and they even run [their own bank](#) to help fund the various non-profit efforts of their club members!

In addition to [donating](#), you can support Hack Club by:

- Starting or joining an in-person Hack Club
- Attending or organizing a hackathon
- Joining their Slack channel or signing up for their newsletter

Other Ideas

- Check if one of your local schools or libraries has a coding club, robotics club, or a makerspace. You could help them purchase supplies, be a guest speaker, become a coach, offer to show them the place where you work, or help judge one of their competitions.
 - FIRST Robotics seems to be a popular student competition. I don't know much about it but I bet they are always in need of coaches or judges!
- Is K-12 not quite your thing?
 - Maybe you could find a local college or university and offer to be a technical advisor for one of their senior engineering design projects or be a guest speaker in one of their classes.
 - Heck, maybe you just find the junior engineer in your office and take them out to lunch. You were that person once, too!

- Check if there are any public makerspaces in your area. You could donate equipment, become a member, or offer to teach a workshop.
- Don't forget to get your employer involved! They could help sponsor an event or a fundraiser; maybe they're just waiting for someone to ask them if they can do it.

Above all else, I would implore you to do *something*. As we enter another season of giving, I ask you to remember the many ways in which you've been blessed in your life and, in whatever way that you can, try to give others those same awesome experiences.

Is there an organization you like to support or another way in which you like to "pay it forward"? Share it with us in the comments!